

LORENZO LAZZARINO

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PHD STUDENT

University of Oxford – STFC



I am a PhD student in Numerical Mathematics at the University of Oxford, advised by Yuji Nakatsukasa and Hussam Al Daas (STFC–CMG). My research concerns (randomized) numerical (multi)linear algebra, with particular interests in randomized low-rank approximations, a-posteriori error estimates, matrix perturbation theory, and the design of synchrotron experiments.

Education

2023–2027	DPhil in Mathematics Advisors: Yuji Nakatsukasa (Oxford) and Hussam Al Daas (STFC–CMG).	University of Oxford, Oxford, UK
2021–2023	MSc in Computational Mathematics Graduated with honours. Thesis: <i>Numerical approximation of the time-ordered exponential for spin dynamic simulation</i> ; supervisor: Stefano Pozza.	Charles University, Prague, Czechia
2021–2023	4EU+ Programme Collaboration with Prof. Christian Bonhomme.	Sorbonne University, Paris, France
2017–2021	BSc in Mathematics	Università Milano–Bicocca, Milan, Italy
2017–2021	Erasmus+ Programme	TU Kaiserslautern, Kaiserslautern, Germany

Publications

- [1] **Blind error estimation for CUR approximation**
L. Lazzarino, K. J. Pearce, N. Pritchard. *arXiv*, 2026. [\[preprint\]](#)
- [2] **Reducing acquisition time and radiation damage: data-driven subsampling for spectro-microscopy**
M. Meier, L. Lazzarino, B. Shustin, H. Al Daas, P. Quinn. *Journal of Synchrotron Radiation*, 2026. [\[paper\]](#)
- [3] **Efficient error estimators for Generalized Nyström**
L. Lazzarino, K. J. Pearce, N. Pritchard. *arXiv*, 2026. [\[preprint\]](#)
- [4] **Preconditioned normal equations for solving discretised partial differential equations**
L. Lazzarino, Y. Nakatsukasa, U. Zerbini. *arXiv*, 2025. [\[preprint\]](#)
- [5] **Matrix perturbation analysis of methods for extracting singular values from approximate singular subspaces**
L. Lazzarino, H. Al Daas, Y. Nakatsukasa. *SIMAX*, 2024. [\[paper\]](#)

Teaching

2024–2026	Tutor, Numerical Linear Algebra Lecturer: Y. Nakatsukasa.	University of Oxford
2026	Mentor, Directed Reading Programme Reading programme based on Pearce and Martinsson (2025).	University of Oxford
2023	Marker, Numerical Linear Algebra Lecturer: J. Tanner.	University of Oxford

2024 **Marker / Teaching Assistant, Continuous Optimization** University of Oxford
Lecturer: C. Cartis.

Awards

2026 **Finalist, STEM for BRITAIN** House of Commons, UK Parliament
Recognised for work on data-driven experimental design in spectro-microscopy.

2023–2027 **Ada Lovelace Centre Scholarship** STFC

2022 **Scholarship for Outstanding Academic Achievements** Charles University

2022–2023 **Student Faculty Grants** Faculty of Mathematics and Physics, Charles University
Team production of Functional Analysis lecture notes in Czech, English, and Italian using a shared mathematical script.

Academic Activities

2026–2027 **Co-organizer, Numerical Analysis Group Internal Seminar** University of Oxford
Co-organizer with Kars Knook.

Talks

04 Jun 2026 **Error estimations for randomized low-rank approximations**
Numerical Analysis Group Internal Seminar | University of Oxford

22 May 2026 **A-posteriori error estimates for randomized low-rank approximations**
ILAS 2026 | Virginia Tech

11 May 2026 **Error estimation for randomized low-rank approximations**
PhD Showcase | RAL Campus

16 Apr 2026 **Error estimation for randomized low-rank approximations**
COMPMATH Seminar | University of Groningen

17 Mar 2026 **Reducing acquisition time and radiation damage through data-driven experimental design in spectro-microscopy**
STEM for BRITAIN | House of Commons, UK Parliament

01 Sep 2025 **Extracting accurate singular values from approximate subspaces**
ENUMATH | Heidelberg

25 Jun 2025 **Extracting accurate singular values from approximate subspaces**
The 30th Biennial Numerical Analysis Conference | University of Strathclyde

05 Jun 2025 **Reducing acquisition time and radiation damage: data-driven subsampling for spectromicroscopy**
Numerical Analysis Group Internal Seminar | University of Oxford

20 Feb 2025 **Matrix perturbation analysis of methods for extracting singular values given approximate subspaces**
Numerical Analysis Seminar | Charles University, Prague (online)

20–21 Jan 2025 **Extracting accurate singular values from approximate subspaces**
Due Giorni di Algebra Lineare Numerica e Applicazioni | Università di Pisa

10 Oct 2024 **Extracting accurate singular values from approximate subspaces**
PYSANUM | Università di Pisa and Scuola Normale Superiore

29 Jul 2024 **Matrix perturbation analysis of methods for extracting singular values**
Gene Golub SIAM Summer School, 3-minute presentations | Quito

05 Mar 2024 **Error bound on singular values approximations by generalized Nyström**
Numerical Analysis Group Internal Seminar | University of Oxford

20–21 Jul 2023 **Numerical approximation of the time-ordered exponential for spin dynamic simulation**
Workshop on Numerical Mathematics | Charles University, Prague

Referee Experience

2026 **SIAM Journal on Scientific Computing**
1 peer review.

2025 **Journal of Mathematical Analysis and Applications**
2 peer reviews.

2024 **Linear Algebra and its Applications**
1 peer review.